Vedant Dinesh Nawghare

Nagpur, Maharashtra | 7709199466 | vedant.nawghare35@gmail.com | https://www.linkedin.com/in/vedant-nawghare-512966320/ | github.com/Vedantnawghare

Education

Symbiosis Institute of Technology, Nagpur 2024 – 2028
Bachelor of Technology in Computer Science and Engineering – Data Science *CGPA: 8.125/10*

Experience

UpToSkills Apr. 2026 – Jul. 2026
AI/ML Intern → Captain, Artificial Intelligence & Machine Learning *Remote*

- Developed and tested backend-oriented AI/ML software across four assignments covering real-time audio buffering, AI output validation, transcript cleaning, and caching.
- Implemented a producer-consumer audio buffer with Python, FastAPI, WebSockets, NumPy, bounded queues, thread synchronization, overflow handling, and monitoring endpoints.
- Built a schema-based AI output validator using FastAPI and JSON Schema with validation testing, consistency scoring, history tracking, and a Streamlit dashboard.
- Implemented transcript preprocessing/normalization and evaluated Cache-Aside and Write-Through strategies using Redis and SQLite with benchmarking; used Docker and Pytest for development and testing.
- Promoted to Captain on June 29, 2026; coordinated 15+ interns through task allocation, progress tracking, technical support, work review, and GitHub pull-request review and code integration.

Research & Patents

- **IEEE Publication – An Agricultural Machine Learning Framework for Ensemble Regressor-Based Crop Price Prediction** — IEEE TEMSMET 2025. Published in IEEE Xplore in 2026; DOI: 10.1109/TEMSMET65536.2025.11467327.
- **Published Patent Application – System and Method for Detecting Artificial Intelligence Generated Voice Using Hybrid Acoustic and Linguistic Analysis** — Indian Patent Application No. 202521130742 A (Jan. 2026); contributed to implementation.
- **Published Patent Application – A Stacking Ensemble Machine Learning System and Method for Agricultural Crop Price Prediction** — Indian Patent Application No. 202621048319 A (Jun. 2026); contributed to implementation.

Projects

AgriChain AI – Smart Harvest & Market Intelligence | *TypeScript, Next.js, Tailwind CSS, OpenWeather API, Agmarknet* GitHub

- Built a solo agricultural decision-support system that combines real weather forecasts, mandi price trends, soil indicators, logistics costs, and spoilage risk to recommend harvest windows, target markets, and preservation actions.
- Engineered a lightweight, explainable rule-based intelligence engine using Agmarknet data and OpenWeather API, with confidence scoring and trend, climate, soil, logistics, and spoilage factors for low-cost rural deployment.
- Implemented multilingual farmer-facing workflows in English, Hindi, and Marathi with a mobile-first interface designed for low-end Android devices.

Patient Vital Monitoring System | *TypeScript, Python, Web Application* GitHub | Live

- Developed the core application for hospital vital monitoring with nurse-side data entry and role-based admin and doctor dashboards for patients, vitals, diagnoses, and clinical records.

Deepfake Voice Detection & Scam Prevention | *Python, Scikit-learn, Librosa, Whisper* GitHub

- Implemented a hybrid acoustic-linguistic pipeline using MFCC and other audio features, Random Forest classification, Whisper/Google Speech Recognition, keyword detection, and risk scoring; reported 94.5% accuracy, 0.963 F1-score, and 0.985 AUC.

Technical Skills

Languages: Python, SQL, C, C++, Java, JavaScript, TypeScript
Web: HTML/CSS, Next.js, Tailwind CSS
Core: Data Structures & Algorithms, Machine Learning, NLP, Ensemble Models
Frameworks & Tools: FastAPI, Streamlit, Redis, SQLite, Docker, Pytest, Git, GitHub, Pandas, NumPy, Matplotlib
Research Models: BERT, DistilBERT, Sentence-BERT, BigBird, Longformer, EmoBERTa, MentalBERT

Achievements & Competitions

- **Adobe University Hackathon (2026):** Cleared the Round 1 online assessment and advanced to Round 2.
- **EncipherX 4.0 CTF – SVPCET, Nagpur (2026):** Advanced to the final round.
- **Cypher Chronicles – IIIT Nagpur (2024):** Advanced to the final round; also participated in multiple hackathons hosted by IITs, BITS, IIIT Nagpur, and SIT Nagpur, reaching offline stages in selected competitions.
- **Certification:** Data Structures and Algorithms using Python – Part 1, Infosys Springboard (2025).